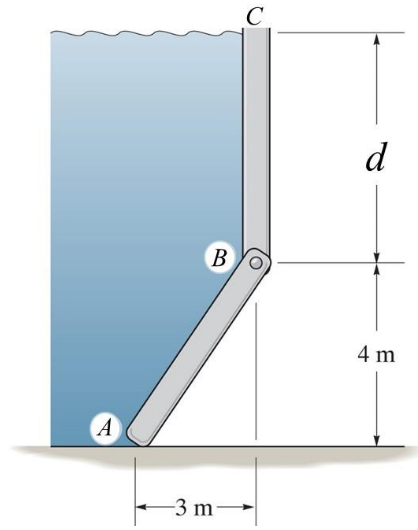
**Practice Exam 4**

1.1 (5 pt) What is the simplest shape that represents the total loading distribution on gate AB ? The gate is 8 m wide.

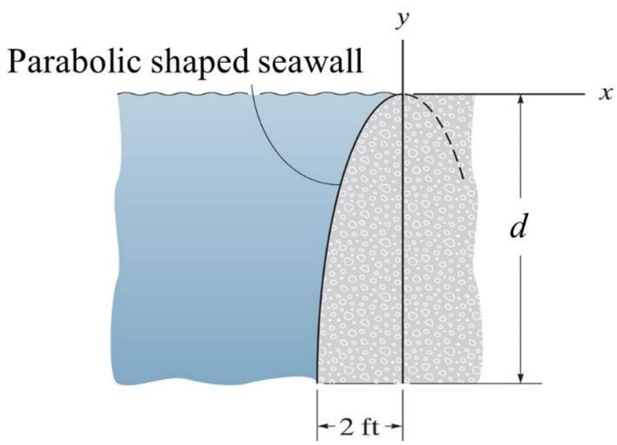



A. trapezoid
B. triangle
C. square
D. rectangle
E. parabolic curve

1.2 (5 pt) How far down from the surface of the water would the center of pressure for the hydrostatic force acting on plate BC be located? d = 6 m

A. 2.67 m
B. 5.33 m
C. 4.00 m
D. 2.00 m
E. 3.00 m

For problems 1.3 and 1.4 - The seawall is shaped in the form of a parabola. The specific weight of water is 62.4 lb/ft3. The wall is 20 ft long and d = 9 ft.




1.3 (5 pt) Determine the magnitude of the horizontal component
of the hydrostatic force acting on the seawall.
.
A. 79.8 kip
B. 50.5 kip
C. 56.2 kip
D. 112 kip
E. 25.3 kip
F. 28.1 kip

1.4 (5 pt) Determine the magnitude of the vertical component of the hydrostatic force acting on the seawall.

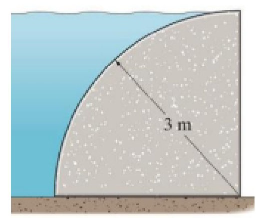
A. 73.5 kip
B. 7.49 kip
C. 15.0 kip
D. 147 kip
E. 3.68 kip

1.2 Check all the **TRUE** statements about fluid pressure problems.

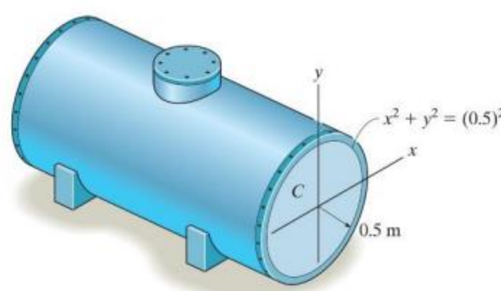
A. The center of pressure is always at the centroid of the submerged plate.
B. The resultant force passes thru the centroid of the loading diagram.
C. The center of pressure is the location where the resultant force strikes the plate.
D. The center of pressure is never at the centroid of the submerged plate.

1.3 What single shape would be used to determine the **Horizontal** component of force on the side of the curved constant width plate submerged in fluid.

1.4 Determine the **Vertical** force component that the water exerts on the side of the quarter circular shaped dam **per meter of length**. Density of water is 1.0 Mg/m³.

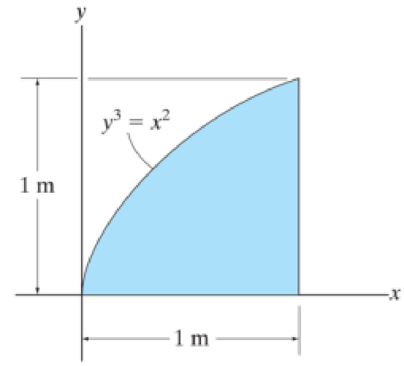



1.5 Determine the **magnitude** of the resultant hydrostatic force acting on the **flat end plate "C"** of the tank. Density of water if 1.0 Mg/m³.



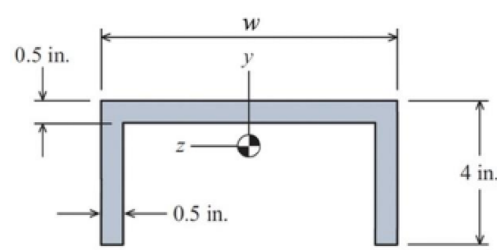
**Section 2 – Moment of Inertia**

2.1 Determine the moment of inertia for the shaded about the x axis.



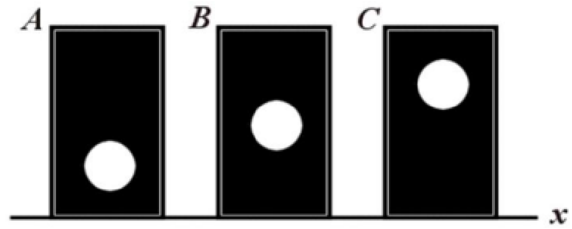

2.2 TRUE FALSE For a general area, the parallel axis theorem would be used when calculating Iy and using a horizontal differential elment.

2.3 If $w$ = 8.50 in., find the moment of inertia about the centroidal $z$-axis. The centroid of the shape is located 2.8468 inches above the bottom of the shape.




2.4 A general shape with an area of 20 in$^2$ has a moment of inertia of 106.7 in$^4$ with respect to the x axis. The centroid is located 2 inches above the x axis. What is the moment of inertia with respect to a centroidal x' axis?

2.5 The figure shows three areas. Each area is a rectangle with a missing circle. The three rectangles have the same dimensions, and the three missing circles have the same dimensions. Which area has the largest moment of inertia about the *x* axis?

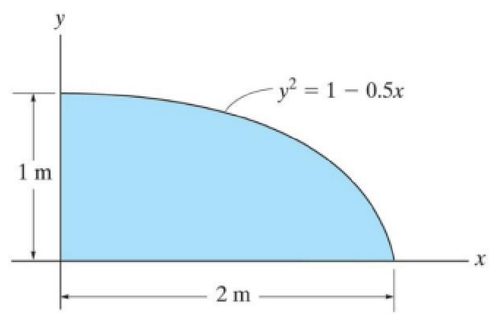



**Section 3 – Product of Inertia**

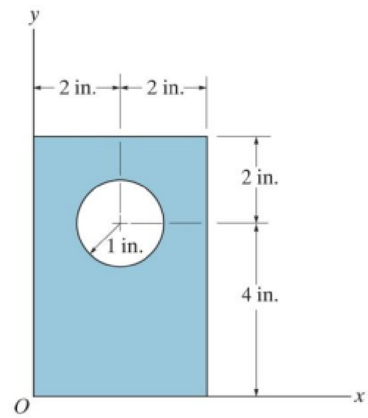
3.1 When using the parallel axis theorem to determine the product of inertia for an area about a set of non-centroidal x-y axes, $I_{xy} = \bar{I}_{x'y'} + Ad_xd_y$, the second term on the right side of the equation is:

A. Area times the differential widths dx and dy
B. Area times the distances between the x-y origin and the x'-y' origin.
C. Always positive
D. Always negative
E. Always zero

3.2 Determine the product of inertia of the shaded area with respect to the *x* and *y* axes.

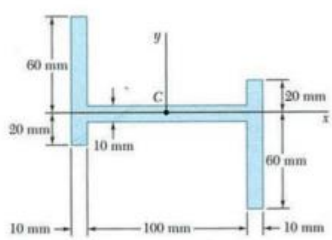


3.3 Determine the product of inertia for the shaded area with respect to the x and y axes.

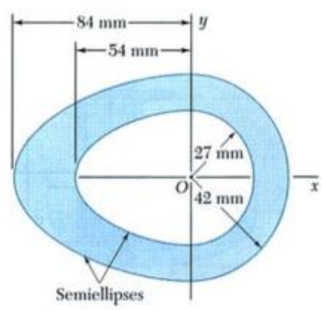



By inspection, determine the sign of Ixy for figures based on the coordinate systems given. Choose from the following choices: positive negative zero

3.4

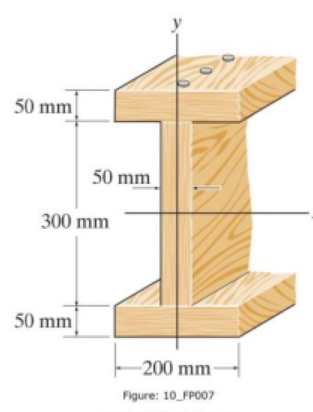



3.5



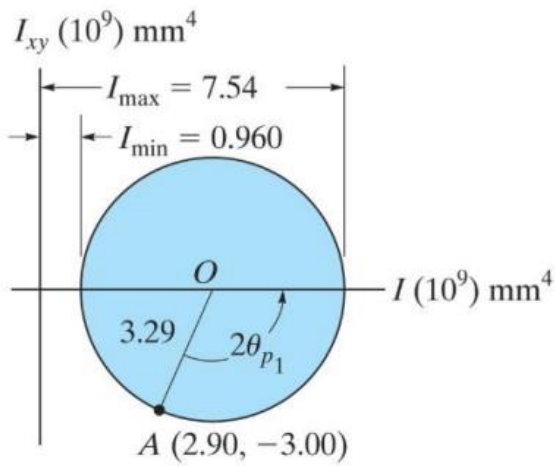

**Section 4 – Inclined Axis and Principle Moments of Inertia**

4.1 The orientation (*relative to the xy axes shown*) of the **principal axes** with origin at the centroid for the cross sectional area would be ______ degrees.



A **Mohr's circle** has the reference point **A** (Ix, Ixy).

The **center** of the circle is at **4.25** and the **radius** is **3.29**




4.2 Iy = _______ *$10^9$ mm$^4$*

4.3 $I_{min}$ = ________ *$10^9$ mm$^4$*

4.4 2θ = ________°

4.5 On the **actual cross section**, the **major** principal axis would be shown rotated _________degrees ________ from the positive x axis.

Answers

1.1 1413 kN

1.2 B and C are true

1.3 Trapezoid

1.4 18.9 kN

1.5 3.85 kN

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2.1 0.111 $m^4$

2.2 TRUE

2.3 11.34 $in^4$

2.4 26.7 $in^4$

2.5 A

3.1 B

3.2 0.333 $m^4$

3.3 119 $in^4$

3.4 negative

3.5 zero

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4.1 0 degrees

4.2 5.6

4.3 .960

4.4 114.2 degrees

4.5 57.1 degrees CCW